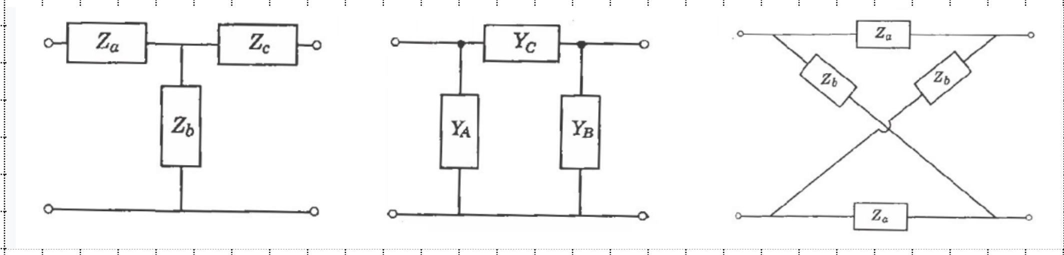
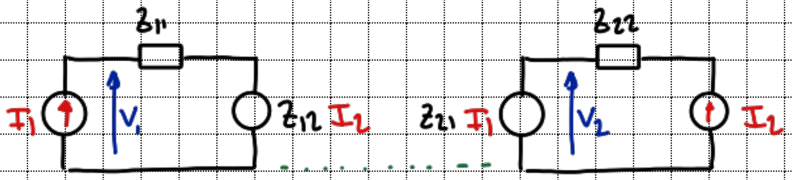
Tarea 6 - Montenegro, Maximiliano



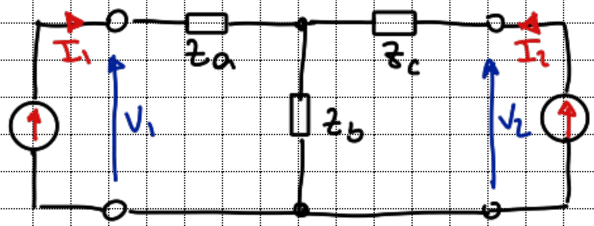
(1) $V = Z\,I$



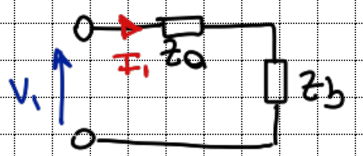
$$\begin{cases} V_1 = z_{11} I_1 + z_{12} I_2 \\ V_2 = z_{21} I_1 + z_{22} I_2 \end{cases}$$

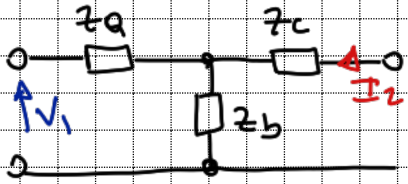
$z_{11} = \left.\frac{V_1}{I_1}\right|_{I_2=0}$ $\quad z_{12} = \left.\frac{V_1}{I_2}\right|_{I_1=0}$

$z_{21} = \left.\frac{V_2}{I_1}\right|_{I_2=0}$ $\quad z_{22} = \left.\frac{V_2}{I_2}\right|_{I_1=0}$



V_1: con $I_2 = 0$ $\qquad$ con $I_1 = 0$

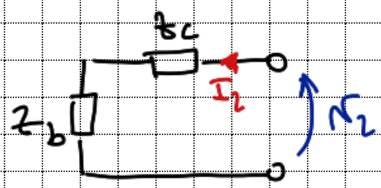


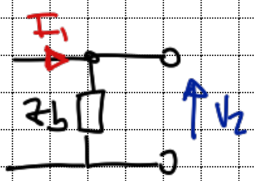


$I_1 = \frac{V_1}{Z_a + Z_b}$ $\qquad$ $I_2 = \frac{V_1}{Z_b}$

$\boxed{\frac{V_1}{I_1} = z_{11} = Z_a + Z_b}$ $\qquad$ $\boxed{\frac{V_1}{I_2} = z_{12} = Z_b}$

V_2 con $I_1 = 0$ $\qquad$ con $I_2 = 0$





$\boxed{\frac{V_2}{I_1} = z_{22} = Z_b + Z_c}$ $\qquad$ $\boxed{\frac{V_2}{I_1} = z_{21} = Z_b}$

$Z_b = Z_{12}$ $\qquad Z_a + Z_b = Z_{11} \Rightarrow Z_a = Z_{11} - Z_{12}$

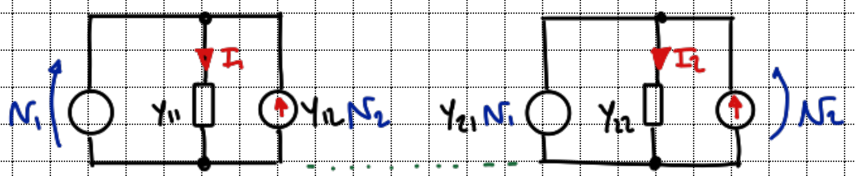
$Z_b + Z_c = Z_{22} \Rightarrow Z_c = Z_{22} - Z_{12}$

$$\begin{cases} V_1 = (Z_a + Z_b) I_1 + Z_b I_2 \\ V_2 = Z_b I_1 + (Z_b + Z_c) I_2 \end{cases}$$

$$\begin{cases} V_1 = (Z_{11} - Z_{12} + Z_{12}) I_1 + Z_{12} I_2 \\ V_2 = Z_{21} I_1 + (Z_{22} - Z_{12} + Z_{12}) I_2 \end{cases}$$

$$\begin{cases} V_1 = Z_{11} I_1 + Z_{12} I_2 \\ V_2 = Z_{21} I_1 + Z_{22} I_2 \end{cases}$$

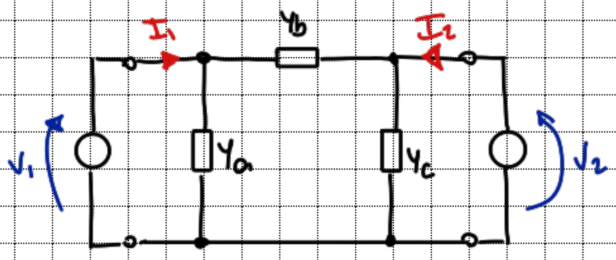
$I = Y V$



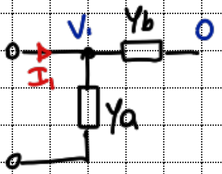
$$\begin{cases} I_1 = Y_{11} V_1 + Y_{12} V_2 \\ I_2 = Y_{21} V_1 + Y_{22} V_2 \end{cases}$$

$Y_{11} = \left.\frac{I_1}{V_1}\right|_{V_2=0}$ $\qquad Y_{12} = \left.\frac{I_1}{V_2}\right|_{V_1=0}$

$Y_{21} = \left.\frac{I_2}{V_1}\right|_{V_2=0}$ $\qquad Y_{22} = \left.\frac{I_2}{V_2}\right|_{V_1=0}$



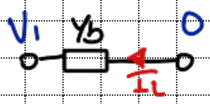
I_1: con $V_2 = 0$



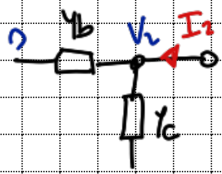
con $V_1 = 0$

$\frac{I_1}{V_1} = \boxed{Y_{11} = Y_a + Y_b}$ $\qquad \frac{I_1}{V_2} = \boxed{Y_{12} = -Y_b}$

I_2: con $V_2 = 0$



con $V_1 = 0$



$$\boxed{\frac{V_1}{I_2} = Y_{21} = -Y_b}$$

$$\boxed{\frac{I_2}{V_2} = Y_{22} = Y_b + Y_c}$$

$Y_{11} = Y_a - Y_{12}$ $\qquad Y_b = -Y_{21} = -Y_{12}$ $\qquad Y_{22} = -Y_{12} + Y_c$

$Y_a = Y_{11} + Y_{12}$ $\qquad Y_c = Y_{12} + Y_{22}$

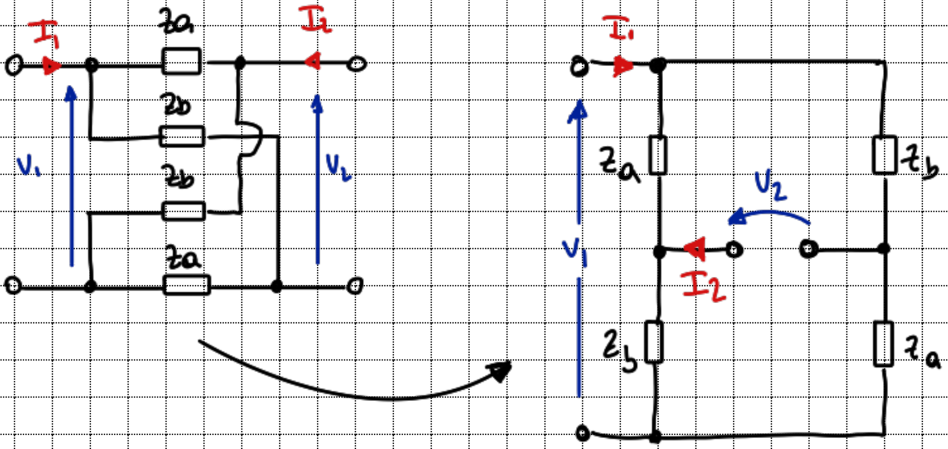
$$\begin{cases} I_1 = (Y_{11} + Y_{12} - Y_{12})\, V_1 + (-Y_{12})\, V_2 \\ I_2 = (-Y_{21})\, V_1 + (Y_{12} + Y_{22} - Y_{12})\, V_2 \end{cases}$$

$$\begin{cases} I_1 = Y_{11} V_1 - Y_{12} V_2 \\ I_2 = -Y_{21} V_1 + Y_{22} V_2 \end{cases}$$

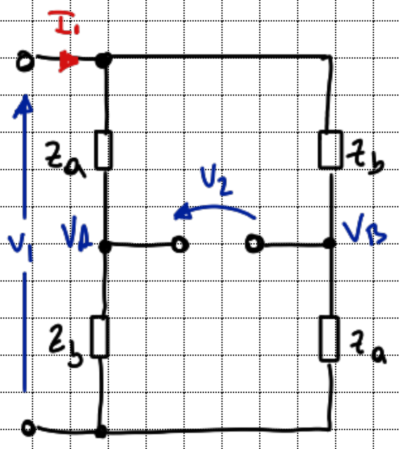
$$\begin{cases} V_1 = V_2\, A + (-I_2)\, B \\ I_1 = V_2\, C + (-I_2)\, D \end{cases}$$

$A = \left.\frac{V_1}{V_2}\right|_{I_2=0}$ $\qquad B = \left.\frac{V_1}{-I_2}\right|_{V_2=0}$

$C = \left.\frac{I_1}{V_2}\right|_{I_2=0}$ $\qquad D = \left.\frac{I_1}{-I_2}\right|_{V_2=0}$



V_1: $i_2 = 0$



$$V_2 = V_A - V_B$$

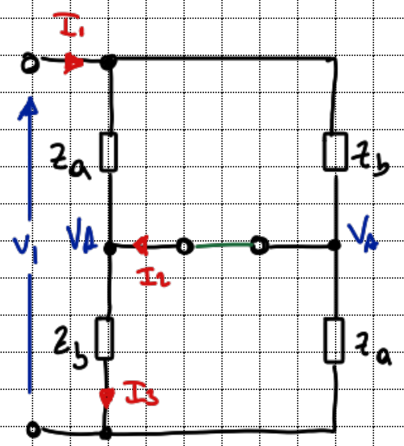
$$V_A = V_1 \frac{Z_b}{Z_a + Z_b}$$

$$V_B = V_1 \frac{Z_a}{Z_a + Z_b}$$

$$\Rightarrow V_2 = V_1 \frac{Z_b}{Z_a + Z_b} - V_1 \frac{Z_a}{Z_a + Z_b}$$

$$\frac{V_2}{V_1} = \frac{1}{A} = \frac{Z_b - Z_a}{Z_a + Z_b} \Rightarrow \boxed{A = \frac{Z_a + Z_b}{Z_b - Z_a}}$$

V_1: $V_2 = 0$



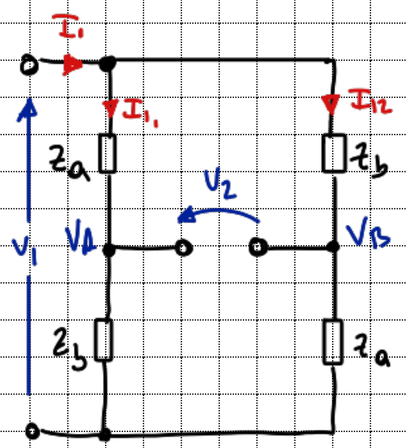
$$I_3 = I_{11} + I_2 \qquad I_{11} = \frac{V_1 - V_A}{Z_a}, \quad V_A = \frac{V_1}{2}$$

$$I_3 = \frac{V_A}{Z_b} \qquad I_2 = I_3 - I_{11}$$

$$I_2 = \frac{V_1}{2Z_b} - \frac{V_1}{Z_a} + \frac{V_1}{2Z_a} \qquad I_2 = V_1 \left(\frac{1}{2Z_b} - \frac{1}{2Z_a} \right)$$

$$I_2 = \frac{V_1}{2} \frac{Z_a - Z_b}{(Z_a Z_b)} \qquad \frac{V_1}{-I_2} = \boxed{B = \frac{2 Z_a Z_b}{Z_b - Z_a}}$$

i_1: $i_2 = 0$



$$I_1 = I_{11} + I_{12}$$

$$I_{11} = \frac{V_1}{Z_a + Z_b} \qquad I_{12} = \frac{V_1}{Z_a + Z_b} \Rightarrow I_{11} = I_{12}$$

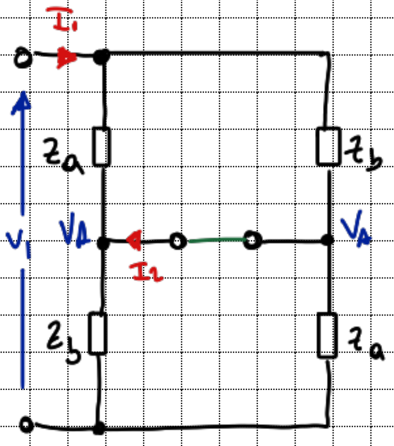
del parámetro B tengo $\frac{I_1}{2} = I_{11}$

$$V_2 = V_1 \frac{Z_b}{Z_a + Z_b} - V_1 \frac{Z_a}{Z_a + Z_b}$$

$$V_2 = \frac{I_1}{2} Z_b - \frac{I_1}{2} Z_a$$

$$V_2 = \frac{I_1}{2} (Z_b - Z_a) \qquad \frac{I_1}{V_2} = \boxed{C = \frac{2}{Z_b - Z_a}}$$

i_1 : $V_2 = 0$



$I_1 = I_{11} + I_{12}$ $\quad I_{11} = \dfrac{V_1 - V_A}{Z_a}$ $\quad I_{12} = \dfrac{V_1 - V_A}{Z_b}$

$$I_1 = \frac{V_1 - V_A}{Z_a} + \frac{V_1 - V_A}{Z_b}$$

$$I_1 = V_1\left(\frac{Z_a + Z_b}{Z_a Z_b}\right) - V_A\left(\frac{Z_a + Z_b}{Z_a Z_b}\right) \qquad V_A = \frac{V_1}{2}$$

$$I_1 = V_1\left(\frac{Z_a + Z_b}{Z_a Z_b}\right) - \frac{V_1}{2}\left(\frac{Z_a + Z_b}{Z_a Z_b}\right)$$

$$I_1 = \frac{V_1}{2}\left(\frac{Z_a + Z_b}{Z_a Z_b}\right)$$

Trayendo de B

$$I_2 = \frac{V_1}{2}\,\frac{Z_a - Z_b}{(Z_a Z_b)}$$

$$V_1 = \frac{2\ Z_a Z_b}{Z_a - Z_b}\, I_2$$

$$I_1 = \frac{\cancel{2 Z_a Z_b}}{Z_a - Z_b} \cdot \frac{1}{\cancel{2}}\left(\frac{Z_a + Z_b}{\cancel{Z_a Z_b}}\right) I_2$$

$$\frac{I_1}{-I_2} = D = \frac{Z_a + Z_b}{Z_b - Z_a} = A$$

$$Z = \begin{pmatrix} Z_a + Z_b & Z_b \\ Z_b & Z_b + Z_c \end{pmatrix} \quad Y = \begin{pmatrix} Y_a + Y_b & -Y_b \\ -Y_b & Y_b + Y_c \end{pmatrix} \quad T = \begin{pmatrix} \dfrac{Z_a + Z_b}{Z_b - Z_a} & \dfrac{2 Z_a Z_b}{Z_b - Z_a} \\ \dfrac{2}{Z_b - Z_a} & \dfrac{Z_a + Z_b}{Z_b - Z_a} \end{pmatrix}$$

(2) $\begin{cases} V_1 = A V_2 - I_2 B \\ I_1 = C V_2 - I_2 D \end{cases}$ $T \longrightarrow Z$

$C V_2 = I_1 + I_2 D$

$V_2 = \frac{I_1 + I_2 D}{C}$

$V_2 = I_1 \frac{1}{C} + I_2 \frac{D}{C}$

$V_1 = \frac{A}{C}(I_1 + I_2 D) - I_2 B$

$V_1 = I_1 \frac{A}{C} + I_2\left(\frac{AD}{C} - B\right)$

$V_1 = I_1 \frac{A}{C} + \frac{I_2}{C}(AD - BC)$

$$\begin{pmatrix} V_1 \\ V_2 \end{pmatrix} = \frac{1}{C}\begin{pmatrix} A & AD-BC \\ 1 & D \end{pmatrix}\begin{pmatrix} I_1 \\ I_2 \end{pmatrix}$$

$V = I \cdot Z$

$$\begin{pmatrix} V_1 \\ V_2 \end{pmatrix} = \begin{pmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{pmatrix}\begin{pmatrix} I_1 \\ I_2 \end{pmatrix} \Rightarrow \quad Z_{11} = \frac{A}{C} \quad Z_{12} = \frac{AD-BC}{C} \quad Z_{21} = \frac{1}{C} \quad Z_{22} = \frac{D}{C}$$

$$T = \begin{pmatrix} \frac{Z_a + Z_b}{Z_b - Z_a} & \frac{2 Z_a Z_b}{Z_b - Z_a} \\ \frac{2}{Z_b - Z_a} & \frac{Z_a + Z_b}{Z_b - Z_a} \end{pmatrix}$$

$$\frac{A = \frac{Z_a + Z_b}{Z_b - Z_a}}{C = \frac{2}{Z_b - Z_a}} \Rightarrow \frac{A}{C} = \frac{Z_a + Z_b}{2} = Z_{11} \qquad \frac{A}{C} = \frac{D}{C} = Z_{22} = \frac{Z_a + Z_b}{2}$$

$$\frac{1}{C} = \frac{1}{\frac{2}{Z_b - Z_a}} = \frac{Z_b - Z_a}{2} = Z_{21}$$

$$A \cdot D - BC = \frac{(Z_a + Z_b)(Z_a + Z_b)}{(Z_b - Z_a)(Z_b - Z_a)} - \frac{2 Z_a Z_b \cdot 2}{(Z_b - Z_a)^2} = \frac{Z_a^2 + 2 Z_a Z_b + Z_b^2 - 4 Z_a Z_b}{(Z_b - Z_a)^2}$$

$$= \frac{Z_a^2 - 2 Z_a Z_b + Z_b^2}{(Z_b - Z_a)^2} = \frac{(Z_b - Z_a)^2}{(Z_b - Z_a)^2} = 1 = AD - BC$$

$$\Rightarrow \frac{AD - BC}{C} = \frac{1}{C} \Rightarrow Z_{21} = \frac{Z_b - Z_a}{2}$$

$$\begin{cases} V_1 = z_{11} I_1 + z_{12} I_2 \\ V_2 = z_{21} I_1 + z_{22} I_2 \end{cases} \qquad z \longrightarrow y \qquad \begin{cases} I_1 = y_{11} V_1 + y_{12} V_2 \\ I_2 = y_{21} V_1 + y_{22} V_2 \end{cases}$$

$$\frac{V_1 - z_{11} I_1}{z_{12}} = I_2$$

$$V_2 = z_{21} I_1 + \frac{z_{22}}{z_{12}} (V_1 - z_{11} I_1)$$

$$V_2 - V_1 \frac{z_{22}}{z_{12}} = I_1 \left(z_{21} - \frac{z_{22} z_{11}}{z_{12}} \right) \Rightarrow z_{12} V_2 - z_{22} V_1 = I_1 (z_{12} z_{21} - z_{11} z_{22})$$

$$I_1 = \frac{1}{z_{12} z_{21} - z_{11} z_{22}} (z_{12} V_2 - z_{22} V_1), \quad z_{12} z_{21} - z_{11} z_{22} = -\Delta z$$

$$I_1 = -\frac{1}{\Delta z} (z_{12} V_2 - z_{22} V_1) \qquad I_1 = \frac{1}{\Delta z} (V_1 z_{22} - V_2 z_{12})$$

$$I_2 = \frac{V_1}{z_{12}} - \frac{z_{11}}{z_{12}} \cdot \frac{1}{\Delta z} (V_1 z_{22} - V_2 z_{12})$$

$$I_2 = \frac{V_1}{z_{12}} - \frac{z_{11}}{z_{12}} \frac{z_{22}}{\Delta z} V_1 + \frac{z_{11}}{z_{12}} \frac{z_{12}}{\Delta z} V_2$$

$$I_2 = V_1 \left(\frac{\Delta z - z_{11} z_{22}}{\Delta z \, z_{12}} \right) + \frac{z_{11} z_{12}}{z_{12} \Delta z} V_2$$

$$I_2 = V_1 \left(\frac{\cancel{z_{11} z_{22}} - z_{12} z_{21} - \cancel{z_{11} z_{22}}}{\Delta z \, z_{12}} \right) V_1 + \frac{z_{11}}{\Delta z} V_2$$

$$I_2 = \frac{1}{\Delta z} (V_2 z_{11} - z_{21} V_1)$$

$$\begin{pmatrix} I_1 \\ I_2 \end{pmatrix} = \frac{1}{\Delta z} \begin{pmatrix} z_{22} & -z_{12} \\ -z_{21} & z_{11} \end{pmatrix} \begin{pmatrix} V_1 \\ V_2 \end{pmatrix}$$

$$y_{11} = \frac{z_{22}}{\Delta z} \qquad y_{12} = \frac{-z_{12}}{\Delta z}$$

$$y_{21} = \frac{-z_{21}}{\Delta z} \qquad y_{22} = \frac{z_{11}}{\Delta z}$$

$$Y \longrightarrow T$$

$$\begin{cases} I_1 = Y_{11} V_1 + Y_{12} V_2 \\ I_2 = Y_{21} V_1 + Y_{22} V_2 \end{cases} \longrightarrow \begin{cases} V_1 = A V_2 - I_2 B \\ I_1 = C V_2 - I_2 D \end{cases}$$

$$\frac{I_2 - Y_{22} V_2}{Y_{21}} = V_1$$

$$I_1 = Y_{11} \frac{1}{Y_{21}} I_2 - Y_{11} \frac{Y_{22}}{Y_{21}} V_2 + Y_{12} V_2$$

$$I_1 = \frac{Y_{12} Y_{21} - Y_{11} Y_{22}}{Y_{21}} V_2 + \frac{Y_{11}}{Y_{21}} I_2 \qquad Y_{11} Y_{21} - Y_{11} Y_{22} = -\Delta Y$$

$$\begin{cases} I_1 = -\frac{\Delta Y}{Y_{21}} V_2 + Y_{11} (-I_2) \\ V_1 = -\frac{Y_{22}}{Y_{21}} V_2 + \frac{1}{Y_{21}} (-I_2) \end{cases}$$

$$\begin{pmatrix} I_1 \\ V_1 \end{pmatrix} = \frac{1}{Y_{21}} \begin{pmatrix} -\Delta Y & Y_{11} \\ -Y_{22} & 1 \end{pmatrix} \begin{pmatrix} V_2 \\ -I_2 \end{pmatrix}$$

$$A = -\frac{\Delta Y}{Y_{21}} \qquad B = -\frac{Y_{11}}{Y_{21}}$$

$$C = \frac{-Y_{22}}{Y_{21}} \qquad D = -\frac{1}{Y_{21}}$$